

Manual

Archiving of Batch / Lot Data TRT-ARC 8.1

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1 Archiving of Batch / Lot Data

Purpose

The function package for Archiving batch/lot data provides access to archived data and, moreover, the possibility to export recorded data and load these data back into the system as needed.

Implementation considerations

You use the function package for archiving batch/lot data, if:

- You request access to data from the Material and Production Logistics as well as the Tracking & Tracing function packages which were already transferred to archive charts.
- You need to ensure the long-term storage of recorded data due to legal or customer requirements.
- You need to import already exported data back into the system for repeated evaluation/analysis due to legal or customer requirements.

Integration

The function package for archiving batch/lot data provides access to inventory data and recorded data of the Material and Production Logistics as well as the Tracking & Tracing function packages.

The function package for archiving batch/lot data provides access to inventory data and recorded data in the relevant archive tables of the Material and Production Logistics as well as the Tracking & Tracing function packages.

Features

- Direct access to archive charts from the application
 - Direct access to archived batch/lot data via the lot data overview and maintenance as well as tabular and graphical batch/lot tracking (prerequisite: you use the TRT-GLV function package)
- Export function
 - Functions for transferring (exporting) data from the archive charts to external file systems for the purpose of long-term storage of the recorded batch/lot data.
- Import function
 - Functions for importing exported data into archive charts for the purpose of evaluating them using the batch data summary and maintenance as well as tabular and graphical batch/lot tracing (prerequisite: you use the TRT-GLV function package).

2 Data Management

Summary

Menu	System Administration → Archiving → Data Management
Transaction code	arccfg
Function authorization	arccfg.*

Utilization

The application has been designed to view or change the centrally managed settings for archiving recorded data.

Integration

The settings for archiving/data management are performed centrally in the application by all components/functions.

Field Descriptions

Product

The HYDRA module, for which a rule is to be defined, has to be entered in the “product” field.

Object

The object defines which kind of data is to be provided.

Retention period

The “retention period” fields define how long data is available until it is archived.

Unit

The unit for retention periods is indicated in days, months or years.

Last retention date

This field is computed and indicates the last day on which data is still available in this object.

Action

It is possible to choose from three variants for one object:

D = delete	object is deleted
M = move (archive)	object is transferred to the next area
X = export	object is unloaded (XUNLOAD format), then deleted

Target object

Automatically results from the detail configuration at the moment.

Condition

The characters of this field are added as another condition to the database command controlling the action. Whereas “and” links this condition with the conditions of the original command and this condition is written in brackets.

Last run (date/time)

Specifies when this rule was carried out the last time.

License

Specifies which license is required for the archiving function. Several licenses can be entered here (separated by blanks). If none of these licenses are available on the system, data are either deleted (action D) or unloaded in files (action M or X).

Path

Optional path for the generation of the file export (unload). The archiving program currently only supports local drives on the HYDRA server. If no path is set file exports are filed by default in:

<HYDRADIR>/<INSTANCE>/custom/archive/<YYYY-MM-DD>/<PRODUCT>/

HYDRADIR ...HYDRA directory

INSTANCE ... Instance number

YYYY-MM-DD Archiving day (YYYY ... year, MM ... month, DD ... day)

PRODUCT ... Product from archiving configuration

Management table

Table name of the management table in which the log records of archiving are saved.

Retention period, management table

The “retention period” fields define how long log records are to be hold in the management table until they are deleted.

Unit

The unit for the retention period is indicated in days, months or years.

Archiving step

Specifies whether this archiving executes the archiving function I or II. Archiving function I moves data from the online dataset to archive tables or deletes them: Setting M (medium-term archive). Archiving function II transfers data from archive tables to the file export or deletes them: setting L (long-term archive).

Configuration

Control indicator saying whether the configuration is active or inactive. Possible values J/N.

Archiving type

Flag for archiving based on time or objects. Supported modes are: O = archiving based on objects (i.e. data are archived separately for each object). Z = archiving based on time (i.e. data are archived together without reference to the object. Please note: Archiving based on time and objects differentiate significantly with respect to the performance (runtime) of the archiving function. We do not recommend archiving mass data on the basis of objects

Priority

Integer value greater than 0. Specifies the execution sequence if several objects are defined for a module. Execution starts with the lowest value configured.

Master table

Table including archiving data. The enhancements entered in the "condition" field refer to this table.

Date column

Date column in the master table. This column is used to evaluate the data to be archived concerning the retention period.

Key 1

Unique key column within the master table by way of which the data to be archived can be identified. Up to 5 key columns can be defined with object-based archiving. Time-related archiving only supports one primary key in key 1.

Key 2 – 5

Further possible key columns for object-based archiving.

Comment

Comment line to describe the archiving configuration.



At the moment only the data management configuration is copied along with copying the archiving configuration. However, defined data records from object details are not included. To do so, the user has to copy the data records of object details to the new configuration using the corresponding dialog.

3 Module-Specific Functions/Configurations

3.1 HYDRA-CAQ

The document entitled [MBL Archiving CAQ.pdf](#) describes the configurations of CAQ 8.1 data management.

3.2 HYDRA-BDE

The document entitled [MBL Archiving BDE.pdf](#) describes the configurations of BDE 8.1 data management.

3.3 HYDRA-MDE

The document entitled [MBL Archiving MDE.pdf](#) describes the configurations of MDE 8.1 data management.

3.4 HYDRA-WRM

3.4.1 Summary

The archiving function makes it possible to archive

- resource documents,
- events as well as
- additional information

according to project-specific, internal or even legal definitions to be still able to access those at a later point in time.

Data considered by HYDRA-WRM archiving are:

- Documents (res_belege table)
- Events (res_event table)

- Additional information (event_dlg_data, hyd_logging, hyd_logging_data table)

Please note

The WRM-ARC license needs to be available to be able to use this archiving function.

3.4.2 Configuration

When transferring data to archive tables, those data are transferred the “retention period” of which has been exceeded (in number of days/months/years → see values in brackets in the following table). Default values, which may be configured individually for each system, are defined for implemented archiving processes. In case the WRM-ARC license is not available, data are deleted, once the configured retention period has expired.

Product	Object	Object designation	Transfer	Default interval (as supplied)
WRM	WRMPRO	Log records/documents	Online dataset → medium-term archive	35 days
WRM	A_WRMPRO	Long-term archiving log records/documents	Medium-term archive → long-term archive	3 years
WRM	EREIGWRM	Events	Online dataset → medium-term archive	35 days
WRM	A_EREIGWRM	Long-term archiving of events	Medium-term archive → long-term archive	3 years
WRM	LOG	HYDRA logging data	Online dataset → medium-term archive	35 days
WRM	A_LOG	Long-term archiving of HYDRA logging data	Medium-term archive → long-term archive	3 years

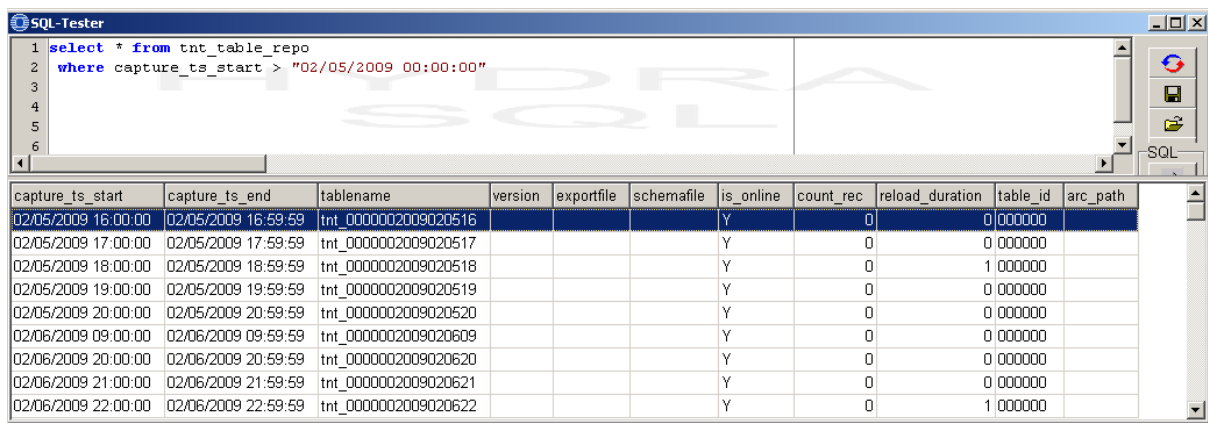
3.5 HYDRA-PDV

3.5.1 Summary

The standard archiving function of HYDRA-PDV 7.2 allows for mass data tables to be archived. This process writes table data in files that are saved in a predefined path.

Mass data tables, also referred to as TNT tables, include a time stamp in their name ID, which is referred to during archiving.

The "tnt_table_repo" table provides information on the archiving status of single TNT tables, e.g. where the data file is located and what's the file's name or whether the table has been re-imported and thus a new export might be required.



The screenshot shows a SQL-Tester window with a query and its results. The query is: `select * from tnt_table_repo where capture_ts_start > "02/05/2009 00:00:00"`. The results table has the following columns: capture_ts_start, capture_ts_end, tablename, version, exportfile, schemafile, is_online, count_rec, reload_duration, table_id, and arc_path. The data rows show various TNT tables with their respective timestamps and IDs.

capture_ts_start	capture_ts_end	tablename	version	exportfile	schemafile	is_online	count_rec	reload_duration	table_id	arc_path
02/05/2009 16:00:00	02/05/2009 16:59:59	tnt_0000002009020516				Y	0		0 000000	
02/05/2009 17:00:00	02/05/2009 17:59:59	tnt_0000002009020517				Y	0		0 000000	
02/05/2009 18:00:00	02/05/2009 18:59:59	tnt_0000002009020518				Y	0		1 000000	
02/05/2009 19:00:00	02/05/2009 19:59:59	tnt_0000002009020519				Y	0		0 000000	
02/05/2009 20:00:00	02/05/2009 20:59:59	tnt_0000002009020520				Y	0		0 000000	
02/06/2009 09:00:00	02/06/2009 09:59:59	tnt_0000002009020609				Y	0		0 000000	
02/06/2009 20:00:00	02/06/2009 20:59:59	tnt_0000002009020620				Y	0		0 000000	
02/06/2009 21:00:00	02/06/2009 21:59:59	tnt_0000002009020621				Y	0		0 000000	
02/06/2009 22:00:00	02/06/2009 22:59:59	tnt_0000002009020622				Y	0		1 000000	

Patch:

The standard HYDRA-PDV 7.2 installation enables archiving of mass data, thus TNT tables.

- DBPATCH PDV_72 (includes the "pdv_setup" table)
- DBPATCH TNT_72 (includes the "tnt_table_repo" table)

License:

License	Product name	Category	Product	Number	Valid until	Release
	PDV	HYDRA	PDV-ARC	1		7.2

3.5.2 Configuration

Archiving of mass data is realized through two components: export and import.

The export represents a cyclic process, which is started by the HYDRA Scheduler and unloads "due" tables from the database in data files on the hard disk.

The import function (or reload) constitutes a library that, if required, reloads data on request from the data files back to the database tables where they can be accessed again.

Normally, the export function transfers the files 1:1 from the database into data files. If configured respectively, however, it is also possible to define a 1:n relation, i.e. a table can be distributed among several files. Vice versa, however, one file always belongs to exactly one table (a table may have several files but a file always belongs to one and only one table).

Program: hp_mexp.exe / out

Installation: The program is entered in the HYDRA Scheduler and started cyclically.

Console menu: File – System administration – Scheduler

The corresponding time, when the program is to be started can be defined in the “fix” tab.

Please note: In the Scheduler the export program should not be started with small intervals, as it is an archiving program. It should suffice to run the program every night or every week depending on the data volume.

Moreover, a parameter has to be defined indicating which conditions are required to consider data to be old enough to be archived. In this case, the parameter refers to the number of tables that may be online. If this number is exceeded the oldest TNT tables are archived by running an export. The parameter can be found in the “pdv_setup” table and may be adjusted in the “PDV” tab of the basic settings of the console.

As part of the implementation process, it has to be checked whether the archiving path exists with ID “PDVARC” in the “arc_path” column of the “pdv_setup” table (and if applicable, whether the transport path is defined as well with the “PDVTRANS” ID in the “trans_path” table).

The corresponding directories which are referred to, can be found in the “hy_path” table.

A host must not be defined if a relative URL path is defined for the archiving function, PDVARC, (subject to the hp_mexp.scr file).

Furthermore, it has to be taken into account that, within the scope of a customer modification, it is possible to archive further data (e.g. events) as well. For this purpose, respective entries have to be defined in the “hyd_datamanagement” table. In this context, the archiving process corresponds to that of the general archiving function.

3.6 HYDRA-MPL / TRT

3.6.1 Summary

By default, the HYDRA-MPL module (product version 7.1 and 7.2) keeps most of the batch data 7 days, before they are moved to the long-term data area.

Different HYDRA-MPL evaluations provide the option to access data that are older than 7 days. For this purpose, HYDRA-MPL postings are put in a special medium-term or archive area. Such data may be accessed by choosing the “consider long-term data” option and entering the required period of time.

Data taken into account by HYDRA-MPL archiving is:

- Batch stock
- Batch attributes
- Batch assignments
- Batch events
- Material movements
- Batch logs (MPL-PRO)

3.6.2 Configuration

Using HYDRA data management, the retention period of data may be configured in the single data areas.

When transferring data into archive tables, such data is taken over the “retention period” of which (number in days/months/years) has been exceeded (see values in brackets in the below table). In case the archiving license for HYDRA-MPL 7.1/7.2 has not been purchased data are deleted after the configured storage period.

Product	Object	Object designation	Transfer	Default interval (as supplied)
MPL71	LOSDELETE	Batches with “deleted”	Online dataset	0 days

Product	Object	Object designation	Transfer	Default interval (as supplied)
MPL72		status	→ deleted	
MPL71 MPL72	MPLBAHNVERT	Cutting plan/cutting plan of operation	Online dataset → deleted	0 days
MPL71 MPL72	LOSAB	Period of time after that processed batches, which are not packed, are deleted/archived.	Online dataset → medium-term archive	7 days
MPL72	LOSPACKED	Period of time after that packed batches are deleted/archived in handling units/merged batches.	Online dataset → medium-term archive	7 days
MPL71 MPL72	A_LOSBESTAND	Archive of batch stock	Medium-term archive → long-term archive	2 years
MPL71 MPL72	LOSEXPIRED	Period of time after that expired batches are deleted/archived.	Online dataset → medium-term archive	7 days
MPL71 MPL72	LOSLEER	Batch without quantity	Online dataset → medium-term archive	7 days
MPL71 MPL72	LOSMATPUF	Batches without material buffer	Online dataset → medium-term archive	7 days
MPL71 MPL72	LOSWASTEBASKET	Batches with material buffer "recycle bin"	Online dataset → medium-term archive	1 day
MPL71 MPL72	LOSTRANSFORT	Batches that have been transferred to an external system.	Online dataset → medium-term archive	0 days
MPL71 MPL72	LOSZUORD	Batch tracing (subject to batch stock)	Online dataset → medium-term archive	-
MPL71 MPL72	A_LOSZUORD	Subject to batch stock	Medium-term archive → long-term archive	See above
MPL71 MPL72	LOSPROTOKOLL	Subject to batch stock	Online dataset → medium-term archive	-
MPL71 MPL72	LOSEVENTMLB	Subject to batch stock	Online dataset → medium-term archive	-
MPL71 MPL72	A_LOSEVENTMLB		Medium-term archive → long-term archive	2 years
MPL71 MPL72	LOSEVENTMLB2	Without batch reference	Online dataset → medium-term archive	7 days
MPL71 MPL72	LOSATTRIBUTE	Subject to batch stock	Online dataset → medium-term archive	0 days
MPL71 MPL72	A_LOSATTRIBUTE	Subject to batch stock	Medium-term archive → long-term archive	See above
MPL71 MPL72	LOSEVENTLOS	Events relating to batches (batch history)	Online dataset → deleted	35 days

3.7 HYDRA-HLS

The document entitled [MBL Archiving HLS.pdf](#) describes the configurations of HLS 8.1 data management.

4 Reload Manager

Summary

Menu	System Administration → Archiving → Reload Manager
Transaction code	arcrlid
Function authorization	arcrlid arcrlid.export arcrlid.import

Utilization

The customer is responsible for backing up data in a data archive and for restoring data. When data is backed up it is transferred from the long-term data area to the archive data area and, as a result, it is stored in a separate file system. The customer can start the backup process using the Reload Manager.

When data, which is filed in the archive data area, is restored, it is transferred back (copied) to the long-term data area. The customer has to do this and bears the responsibility.

To be able to evaluate restored data using standard evaluations/reports, the data has to be transferred back (copied) to the reload data area.

For this reason, the Reload Manager enables the following functions:

1. Moving of exported data into the customer archive (the customer archive path has to be defined within the HYDRA path settings).
2. Loading of exported data into the reload area for evaluations

Integration

The Reload Manager is a central function that is used by many components or functions.

Selection Criteria

The application provides the following selection criteria:

Module

Reference to the product group which archived data belongs to.

Object type

Object type of the archived data.



The selection options “module” and “object type” are mandatory fields.

Toolbar

Export

The “EXPORT” button allows for the entries selected in the Reload Manager to be exported to the customer directory (all data records can be selected using the context menu of the right mouse button). Once the function has been started, the input dialog that opens requires the customer archive path to be entered.

Import

The “IMPORT” button allows for the selected entries to be loaded into the reload area, in order for the data to be again available for HYDRA evaluations/reports. Once the function has been started, the data loading mode and an optional path are requested. By indicating the path, it is possible to specify another storage location for the files, provided that the archived files have not been moved to the customer-specific archive using the Reload Manager.

There are three different ways to deal with reload data after they have been reloaded to prevent the reload data set from increasing excessively (→ no slow evaluations/reports):

The following modes are distinguished:

- **Cyclic**
Data is loaded to the reload area in the “cyclic” mode. Loaded data is automatically removed from the corresponding reload tables, once the retention period specified within archive settings has expired.
Please compare the configuration entry HYD / RELOADMANAGER of the data management function. When demo settings are used, data is removed from the reload area after 14 days.

When data is imported, the HYD_REL_MANAGEMENT.DELETE_DATE field is calculated subject to the configuration and entered in the reload management table.

- **User-specific**
Data is loaded to the reload area in the “user-specific” mode. The loaded data is automatically removed from the corresponding reload area, once the time specified for the user within the user-specific settings has expired. Exception: In case several users loaded identical data into the reload area, the corresponding retention period is automatically set to the maximum date.
When data is imported, the HYD_REL_MANAGEMENT.DELETE_DATE field is computed subject to the configuration and entered in the reload management table.

- Manual

Data is loaded in the “manual” mode to the reload area. The customer is responsible for deleting data from corresponding reload tables. However, identical data cannot be loaded several times in the “manual” mode (data can be loaded in the “user-specific” mode though).

In order for data to be transferred to the reload data area, the HYDRA server must be able to access these files. Otherwise, loading is cancelled with an error message.